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Explaining the Relativistic Nature of General and Special Relativity

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ABSTRACT: In this paper, we reveal that the time dilation between two systems is equal to the ratio of their respective DeGerlia inertial density D, defined as mass over radius. Fundamental to this exploration is the recognition that relativity is always relativistic, meaning there is no single privileged frame of observation: no measurement, whether of time or anything else, has any meaning without a comparator. We reveal this by taking our relative inertial density equation, plugging in a system compared to that same system at its Schwarzschild radius, and demonstrating that we obtain the same exact result as the gravitational time dilation equation. In order to present this equivalence and demonstrate the relativistic principle, we utilize a framework developed by the author called the DeGerlia Time Dilation framework, a simple, mathematically equivalent time dilation framework that is directly derived from general relativity.

Keywords: time dilation, general relativity, special relativity, gravitational time dilation, Lorentz time dilation, DeGerlia inertial density, inertial density, moment of inertia, Schwarzschild radius

1 Introduction

General relativity[1] is a model for observing our universe that is fundamentally comparative: no quantity has physical meaning without a comparator. The pace of time is no exception: the time dilation between two systems is a product of the systems' respective geometric mass distribution. Directly from the gravitational time dilation formula one can derive the underlying two-system logic and ultimately GTD is simply a comparison between a system's pace of time relative to the pace of time of an arbitrarily selected benchmark system, which in this case is the same system at its Schwarzschild radius. The formula allows you the freedom to only insert one mass and one radius, and the second is derived directly from the first. This has the effect of hiding the true nature of relativity and that is that it is, above all, comparative. But sends a signal that one need not compare two systems to understand time, and that is flawed thinking; it introduces the mathematical fallacy of an arbitrarily selected comparator, when there is, in fact, no privileged perspective from which to understand time. In this paper we evaluate the physical property responsible for relativistic time dilation and derive the underlying formulation that quantifies the relative time dilation between two systems.

The relative flow of time between two systems is dictated by their relative DeGerlia inertial density D. In this paper we utilize a principle developed by the author, D, which is an abstracted measure explained herein, that retains the proper unit of this property and that is mass over radius. Thus D is defined as a system's mass over radius[2], $D = M/R$. We do a comparative analysis and tabulate relative time dilation across a variety of systems from classical to relativistic regimes, and show that the same D ratio $k = D_1/D_2 = M_1R_2/(M_2R_1)$ returns the Schwarzschild time-dilation ratio exactly across every tested constraint class, including the fully general case. This establishes that the relative rate of time between systems is not an ad hoc relativistic output requiring separate formulas, but a direct consequence of comparative mass-radius geometry. The result makes explicit a simple relation already contained in general relativity: given two systems' masses and radii, their relative time dilation is determined by their respective geometric distribution of mass alone.

Several distinct, common forms of time dilation utilize the same formula, and that is that the time dilation is equal to $\sqrt{1-k}$. In general relativity, the scale factor is calculated by r_s/R. In special relativity[3], it's calculated as v^2/c^2, and in general relativity, you can also substitute that: r_s/R could be replaced with D/D_{crit}. Those are all equivalent.

Both General and Special Relativity follow a very similar form, and their distinction mirrors the distinction between linear and rotational inertia. This alludes to a very rational and logical unification of relativity.

Outline. General relativity tells us that the time dilation is fundamentally a comparison of the pace of time between two different systems. In the case of the well-known gravitational time di-

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lation formula, $d\tau/dt = \sqrt{1-r_s/R}$. But if you look at the formula deeper it comes out that you're comparing this system of identical mass over the radius that is equivalent to its Schwarzschild radius.

2 Experimental Model

The experimental model employed in this study rests on several key components. The framework is derived directly from general relativity and is therefore mechanically and mathematically equivalent to well-established general-relativistic calculations. It does not rely upon, nor attempt to introduce, any principles not already established within the theories of general and special relativity; it is simply a re-framing of the existing theory of general relativity for exploratory purposes.

We begin by introducing the DeGerlia Framework and then demonstrate its complete mathematical equivalence with general relativity. We then work through a series of examples, each one demonstrating a different comparative case, and in each case we validate the result using standard gravitational time dilation calculations.

The first example is the most familiar: the case in which both systems have the same mass. Gravitational time dilation is such a case, because to calculate it one must compare a system with an identical system at its Schwarzschild radius. This example demonstrates the two-system nature of all relativistic calculations by comparing two systems of equal mass. Relativity is fundamentally concerned with how quantities appear relative to one another; gravitational time dilation reflects the comparison of systems of equal mass, where the compared radius is the collapse-condition radius r_s. We then work through a second equal-mass calculation, comparing an arbitrarily large system to a system of equal mass at four-thirds the Schwarzschild radius, or $d\tau/dt = 0.5$. We work through two systems of equal radius, which represent the Lorentz factor in special relativity. We also perform calculations for systems of equal density or equal moment of inertia, in addition to the classical case in which we compare near-scale objects, and a general case with no constraint, in which one may identify the relative pace of time between any two systems bearing no similarity to one another.

3 Revisiting Key Concepts in Physics

3.1 Inertia and Moment of Inertia

Inertia dictates how a system's motion changes in response to an applied force. This applies to both linear and rotational motion. In rotational dynamics, the "moment of inertia"[4] I dictates how the angular motion of a system about a specific axis changes when a torque is applied, and is defined as the sum of each mass element multiplied by the square of its distance from the axis of rotation.

$$\text{Moment of Inertia: } I = \sum_i m_i r_i^2 \tag{1}$$

Using Equation (1), one can determine how motion changes for any system when energy is applied about an axis (torque). As it pertains to linear motion, inertia (I) is characterized solely by the system's mass.

3.2 General Relativity and Time Dilation

General Relativity defines the formula to calculate a system's time dilation:

$$d\tau/dt = \sqrt{1-k} \tag{2}$$

where $k = r_s/R = 2GM/(Rc^2)$, M is the system's mass, R its radius, $r_s = 2GM/c^2$ its Schwarzschild radius, c the speed of light $= 2.99792458\text{e}8\,\mathrm{m\,s^{-1}}$, and G the gravitational constant, $G = 6.67430\text{e}{-11} \pm 1.5\text{e}{-15}\,\mathrm{m^3\,kg^{-1}\,s^{-2}}$.

The equation requires only a single system's mass and radius, yet k is itself a ratio of two radii: the system's radius R and the Schwarzschild radius r_s of that same mass. The calculation is therefore a comparison of two systems — the system as given, and the same mass at its Schwarzschild radius.

4 The DeGerlia Time Dilation Framework

4.1 Why the DeGerlia Time Dilation Framework

The framework described herein was developed out of necessity. The author spent a considerable amount of time running Schwarzschild equations, calculating Schwarzschild radii, and eventually realized that there was a simpler way to represent some of these things. First and foremost, it is a simpler representation, because we represent the Schwarzschild threshold — the black hole condition — not as a value that has to be calculated for every mass, but as a universal constant that applies to any mass. To cite an example, say you have a system with a mass of 1e30 and a radius of 1e10. All you need to do is take the order-of-magnitude difference between those two exponents. Because $D_{crit} \approx 6.73295\text{e}26$, if that difference is 27 or more it is guaranteed to be a black hole; if it is 25 or less it is guaranteed not to be; and if it falls at exactly 26 it may or may not be a black hole, and you should calculate further. So look at those two numbers: 30 minus 10 is 20. That is not yet a 26; it is not a black hole. It does not impart any risk of imaginary numbers as output. In fact, it accommodates for the actual system behind the event horizon. It does not artificially normalize, shoehorning the concept of time dilation into a range between 0 and 1 which reinforces the notion that flat space time actually exists when in actuality it is an asymptotic approach. It does not conflate the Schwarzschild radius with time stoppage, which would occur at the singularity that is radius zero, however, it is also in asymptotic approach. It calibrates r_s at a value of 1, and anything above that is behavior beyond the threshold — behavior beyond the Schwarzschild radius. It represents the Pythagorean complement to gravitational time dilation, and is therefore completely recoverable as gravitational time dilation and completely consistent with gravitational time dilation and general relativity, with $DTD^2 + GTD^2 = 1$. It does not obfuscate the relative nature of time by providing single system interpretations of time dilation. It allows you to work with numbers that are not on the fringe of available precision, meaning you don't have to work with numbers extremely close to one, like $0.999\ldots$ and so on. In all but the most extreme relativity cases, time dilation is a very small factor across the bulk of the region in which it would be measured. Only in extreme cases do you see values that reach something like 0.5; at four-thirds the Schwarzschild radius a system is at half time, so it falls off extremely fast. This precision is essential because we need these calculations to work for applications like GPS satellites, the International Space Station, and more complex communications. This framework allows you to work across a range of numbers spanning approaching zero to approaching infinity, with all the precision you might need over the entire continuum.

4.2 Definitions used within the DeGerlia Time Dilation Framework

Symbol	Definition / Value
c	Light Speed $299\,792\,458\,\mathrm{m\,s^{-1}}$
D	DeGerlia Inertial Density M/R_{axis}
D_{crit}	DeGerlia Threshold $c^2/2G = 6.732\,95\mathrm{e}26\,\mathrm{kg\,m^{-1}}$
DTD	DeGerlia Time Dilation DTD$=\sqrt{k}$
G	Gravitational Constant $6.674\,30\mathrm{e}{-11} \pm 1.5\mathrm{e}{-15}\,\mathrm{m^3\,kg^{-1}\,s^{-2}}$
GTD	Gravitational Time Dilation $d\tau/dt = \sqrt{1-k}$
k	Linear Scale Factor $v^2/c^2 = r_s/R = D/D_{crit}$
k_D	Linear Scale Factor (DeGerlia inertial density) D/D_{crit}
k_s	Linear Scale Factor (Schwarzschild) r_s/R
k_v	Linear Scale Factor (velocity) v^2/c^2
P	Inertial Density $I/V = D \cdot 3/10\pi$
r_s	Schwarzschild Radius $2GM/c^2$
STE	Space-time Equivalence[5] $k_D = M_1R_2/M_2R_1$

4.3 System

Under the DeGerlia Time Dilation Framework a "system" is defined as "any discrete collection of particles and the space between them", or alternatively, "any unit of space and the particles contained within." Because this framework sees the universe not as a clean separation of discrete material systems separated by the vacuum of space, but as a gradient of material density pervasive everywhere throughout the universe. The most extreme densities reside not within near-scale systems but rather, within the cores of highly dense gravitational objects far from our physical scale, such as black holes or sub-atomic particles, and forms a gradient all the way to the near-scale "vacuum of space" where another gravitational source begins to exert more influence. It does not concern itself with notions of uniformity or even symmetry at all. Sphericalness relies on none of those. It doesn't rely on a clean edge separating matter from vacuum, or anything along those lines. It doesn't rely on static density nor any particular density gradient. Under the DeGerlia framework, the system requires only a mass and a radius about some axis. One can simply say my system consists of the Earth to the mean radius of the surface. But if you were talking about Jupiter, which lacks without a clear periphery separating the surface from the atmosphere, one might characterize the system as out to a radius of x or out to a density of y. If analyzing a large system, it would not be uncommon to take a spherical sample from within that system in order to characterize the behavior of certain regions. It accurately and precisely recovers the gravitational time dilation figures for every pair of systems that can be compared, as shown in Tables 1 and 2.

4.4 System Time

For every whole system — and this applies to the whole system, not to any subsystem — the pace of time is static, and serves as the benchmark for the pace of time of other systems from the perspective of that system. A subsystem will have very different physical properties, and therefore behaves very differently than the system. In general relativity, time dilation is dilated exactly by the mass-to-radius ratio. This means you cannot expect part of a system to have the same pace of time as the whole system. Also, because the underlying relationship is one of inertia and density simultaneously, the inertia component imparts dimensionality to time. As such, despite this exploration focusing on uniform spherical systems, this framework accommodates asymmetrical systems, such as large low-density black holes, Kerr black holes, and even the observable universe.

4.5 Inertial Density P

Inertial density P as shown in (4) is a fundamental system property[2] characterizing the distribution of inertia across any system, whether in the linear or rotational context, in any direction, or about any axis. If we were to redefine classical density as I/v (inertia/volume), because inertia is mass for one-dimensional motion, this would remain consistent with the classical definition of density, where $\rho = m/v$.

Inertial density could be best described as the hollowness, or foaminess, of a system. Consider the object that has the most moment of inertia for its given volume: a shell, an object where all of its mass resides at its radius, as far out as it can physically be placed, concentrated right at the outer edge. Bubbles have this character. They do not jostle easily; they are readily carried by the wind, but it is hard for them to change trajectory, because, per unit volume, they have a great deal of inertia despite being extremely light. That is the highest-inertia structure there is. Conversely, an object with a large radius but all of its mass concentrated at an almost point-like mass at the center gives the opposite: as that mass concentrates infinitely toward the center, the inertia approaches zero. Somewhere in between is a solid sphere. If you were to fill that sphere with small empty spheres, you would distribute more of the mass across the broader structure, so the foamy sphere remains a high-inertia object for its volume.

4.6 Mathematics of Inertial Density

To derive the inertial density relationship, we begin with the established relationship between mass and density ρ as shown in (3):

$$\rho = \frac{m}{v} \tag{3}$$

where ρ is density, m is mass, and v is volume. Because mass is equivalent to linear inertia, and because density is m/v, we define inertial density P in terms of moment of inertia over volume:

$$\boxed{\text{Inertial Density}: P = \frac{I}{v}} \tag{4}$$

where inertial density P is equal to inertia I over volume v. Note, the moment of inertia I for a system is distinct about every axis for systems that are not uniform.
For a uniform sphere[6], we substitute $I = \frac{2}{5}mr^2$ and $v = \frac{4}{3}\pi r^3$:

$$P = \frac{I}{v} = \frac{\frac{2}{5}mr^2}{\frac{4}{3}\pi r^3} = \frac{3m}{10\pi r} \tag{5}$$

Which can be simplified as follows:

$$P = \frac{m}{r} \times \frac{3}{10\pi} \tag{6}$$
$$P = \frac{m}{r} \times 9.549\,296\,585\text{e}{-2} \tag{7}$$

Therefore, for spherical systems, P can be calculated from the ratio of mass to radius multiplied by the conversion factor $\frac{3}{10\pi}$ as shown in (7).

4.7 DeGerlia Inertial Density D

For all spherical systems, m/r multiplied by the geometric factor $3/(10\pi)$ equals inertial density. ***DeGerlia Inertial Density*** is a simplified analog to inertial density I/v that is equal to mass over radius m/r while omitting the $3/10\pi$ geometric conversion factor:

$$\boxed{\text{DeGerlia Inertial Density: } D = \frac{m}{r}} \tag{8}$$

DeGerlia inertial density (8) $D = m/r$ is related to inertial density P by the geometric factor $3/10\pi$ as shown in (9):

$$P = \frac{3}{10\pi} \times D \tag{9}$$

Inertia, in a linear context, is simply mass. Inertial density, in this context, reduces directly to mass density $\rho = M/V$. From this perspective, inertial density is simply the rotational analog to mass density, not unlike the relationship between $F = ma$ and $\tau = I\alpha$. More succinctly stated: mass density is simply the linear interpretation of inertial density.

Just as ordinary density ($\rho = M/V$) is linear inertia (mass) distributed over volume, DeGerlia inertial density is rotational inertia distributed over volume. As stated in the prior work: *mass and density are to linear motion as moment of inertia and inertial density are to rotational motion.*

$D = M/R$ is a simplified analog to inertial density, related to P by the geometric factor $3/10\pi$, and represents the mass over radius ratio for a system. Given that this quantity is almost always used in a ratio that would cancel out all geometric vectors and constants, for ease of use, this metric D for inertial density is typically used over the geometrically accurate inertial density, P.

4.8 DeGerlia Threshold D_{crit}

The Schwarzschild condition can be expressed as a static universal constant threshold of DeGerlia inertial density D. There are several advantages to this approach, not the least of which is that you don't have to calculate the threshold for every mass. And, it's much easier to calculate D than P. Inertial density is an intuitive metric that's easier to work with.

Setting $R = r_s$ (the Schwarzschild condition), the DeGerlia inertial density D becomes the threshold D_{crit}:

$$D_{crit} = \frac{m}{r_s} \tag{10}$$

Recall the Schwarzschild radius[7]:

$$r_s = \frac{2Gm}{c^2} \tag{11}$$

Substitute r_s into the expression for D_{crit}:

$$D_{crit} = \frac{m}{\frac{2Gm}{c^2}} = \frac{mc^2}{2Gm} = \frac{c^2}{2G} \tag{12}$$

Insert the fundamental constants[8], $c = 2.997\,924\,58\text{e}8\,\text{m}\,\text{s}^{-1}$ and $G = 6.674\,30\text{e}{-11} \pm 1.5\text{e}{-15}\,\text{m}^3\,\text{kg}^{-1}\,\text{s}^{-2}$:

$$\begin{aligned} D_{crit} &= \frac{(2.997\,924\,58\text{e}8)^2}{2 \times 6.674\,30\text{e}{-11}}\,\text{kg}\,\text{m}^{-1} \\ &= \frac{8.987\,551\,79\text{e}16}{1.334\,860\text{e}{-10}} \\ &= 6.732\,95\text{e}26\,\text{kg}\,\text{m}^{-1} \end{aligned}$$

$$\boxed{\text{DeGerlia Threshold: } D_{crit} = 6.732\,95\text{e}26\,\text{kg}\,\text{m}^{-1}} \tag{13}$$

This is the universal constant of DeGerlia inertial density at which the escape velocity equals the speed of light. Note this value is limited by the precision of the gravitational constant[9], $G = 6.674\,30\text{e}{-11} \pm 1.5\text{e}{-15}\,\text{m}^3\,\text{kg}^{-1}\,\text{s}^{-2}$. The DeGerlia Threshold represents the mass over radius threshold at which gravitational collapse occurs. Just as the Schwarzschild radius, the DeGerlia threshold is universally applicable to spherical gravitational systems. Examples include black holes or neutron stars, a spherical region of an interstellar cloud, or even a galactic cluster. As defined in (13), this universal threshold provides a simple criterion for determining when a system will undergo gravitational collapse (collapse of the neutron structure).

4.9 DeGerlia Time Dilation

DeGerlia Time Dilation is the geometric complement of gravitational time dilation:

$$DTD = \sqrt{k} \tag{14}$$

where k is the scale factor of Equation 15. Where GTD = $\sqrt{1-k}$ is the surviving proper-time rate $d\tau/dt$, DTD = $\sqrt{k}$ is its complement.

4.10 The Linear Scale Factor k

The following identity relates the time-dilation ratio of two uniform spherical systems to their mass and radius.

$$k = \frac{I_1}{I_2}\left(\frac{R_2}{R_1}\right)^3 = \frac{D_1}{D_2} = \frac{M_1 R_2}{M_2 R_1} \tag{15}$$

where R_1, R_2 are the radii, M_1, M_2 are the masses, and $D_i = M_i/R_i$ is the DeGerlia inertial density of system i. The three forms in Equation 15 are algebraically identical; $(I_1/I_2)(R_2/R_1)^3$ exposes the inertia-radius structure, D_1/D_2 is the DeGerlia inertial density D ratio, and $M_1 R_2/(M_2 R_1)$ is the fully expanded form. A system's pace of time has meaning only relative to another pace of time. The scale factor k governs the relative time-dilation between two uniform spherical systems. The standard gravitational time-dilation calculation takes the Schwarzschild form $\sqrt{1-k}$, where $k = D/D_{crit} = r_s/R$. The D ratio $k = D_1/D_2$ and the Schwarzschild form give the same value. Via the Schwarzschild radius, the same scale factor is:

$$k = \frac{D}{D_{crit}} = \frac{r_s}{R} \tag{16}$$

where $D = M/R$ is the system's DeGerlia inertial density and $r_s = 2GM/c^2$ is its Schwarzschild radius. This is the k that appears in the standard $\sqrt{1-k}$ for GTD.

Time dilation is always:

$$\frac{d\tau}{dt} = \sqrt{1-k} \tag{17}$$

where $d\tau/dt$ is the time-dilation factor between the two systems and k is the DeGerlia inertial density ratio for the case being computed. The specific forms are:

- $k = D_1/D_2$ (Equation 15) — via inertial density ratio
- $k = D/D_{crit} = r_s/R$ (Equation 16) — via Schwarzschild radius

Same k symbol across forms: a given numerical k produces the same time-scale-factor result regardless of which form it appears in.

4.11 Properties of DTD

Completely analogous to gravitational time dilation and part of the exact same general relativity. To translate GTD to DTD, one can leverage the Pythagorean relationship between the two.

GTD and DTD are bound by the Pythagorean identity:

$$GTD^2 + DTD^2 = (1-k) + k = 1. \tag{18}$$

5 Relative Time Dilation Examples

Tables 1 and 2 present the complete system properties and derived ratios for seven system pairs spanning the cases that follow: static mass at the Schwarzschild radius ($M = M$, $R \approx r_s$), static mass at four-thirds the Schwarzschild radius ($M = M$, $R = \frac{4}{3} r_s$), static radius ($R = R$), constant density ($\rho = \rho$), constant inertia ($I = I$), an unconstrained general case, and a classical case. Across every case, the three algebraic forms of the scale factor k (Equation 15) produce identical values, and the time-dilation ratio computed via the DeGerlia route agrees with the gravitational time-dilation ratio computed independently from the standard Schwarzschild form — to the precision afforded by G. No case requires a separate formula. The worked examples in the subsections that follow show, case by case, exactly how each column of these tables is obtained.

Table 1 System properties for each pair comparison case.

Property	Case 1 $M=M$, $\sim r_s$	Case 2 $M=M$, $\sim \frac{4}{3} r_s$	Case 3 $R=R$	Case 4 $\rho=\rho$	Case 5 $I=I$	Case 6 General	Case 7 Classical
M_1 (kg)	2.00000e+30	1.00000e+30	1.00000e+30	1.00000e+30	1.00000e+32	1.00000e+20	8.00000e+0
M_2 (kg)	2.00000e+30	1.00000e+30	1.00000e+24	1.00000e+27	1.00000e+30	1.00000e+30	1.00000e+0
R_1 (m)	2.97049e+3	1.98031e+3	1.00000e+8	1.00000e+8	1.00000e+7	2.00000e-7	2.00000e+0
R_2 (m)	7.00000e+8	1.00000e+8	1.00000e+8	1.00000e+7	1.00000e+8	1.00000e+7	1.00000e+0
ρ_1 (kg/m^3)	1.82161e+19	3.07406e+19	2.38732e+5	2.38732e+5	2.38732e+10	2.98416e+39	2.38732e-1
ρ_2 (kg/m^3)	1.39203e+3	2.38732e+5	2.38732e-1	2.38732e+5	2.38732e+5	2.38732e+8	2.38732e-1
I_1 (kg·m^2)	7.05907e+36	1.56865e+36	4.00000e+45	4.00000e+45	4.00000e+45	1.60000e+6	1.28000e+1
I_2 (kg·m^2)	3.92000e+47	4.00000e+45	4.00000e+39	4.00000e+40	4.00000e+45	4.00000e+43	4.00000e-1
D_1 (kg/m)	6.73289e+26	5.04972e+26	1.00000e+22	1.00000e+22	1.00000e+25	5.00000e+26	4.00000e+0
D_2 (kg/m)	2.85714e+21	1.00000e+22	1.00000e+16	1.00000e+20	1.00000e+22	1.00000e+23	1.00000e+0
r_{s1} (m)	2.97046e+3	1.48523e+3	1.48523e+3	1.48523e+3	1.48523e+5	1.48523e-7	1.18819e-26
r_{s2} (m)	2.97046e+3	1.48523e+3	1.48523e-3	1.48523e+0	1.48523e+3	1.48523e+3	1.48523e-27
$D_{\mathrm{norm},1}$	9.99991e-1	7.50001e-1	1.48523e-5	1.48523e-5	1.48523e-2	7.42617e-1	5.94093e-27
$D_{\mathrm{norm},2}$	4.24352e-6	1.48523e-5	1.48523e-11	1.48523e-7	1.48523e-5	1.48523e-4	1.48523e-27
k_s	9.99990e-1	7.50000e-1	1.48523e-5	1.48523e-5	1.48523e-2	7.42616e-1	5.94093e-27
k_d	9.99991e-1	7.50001e-1	1.48523e-5	1.48523e-5	1.48523e-2	7.42617e-1	5.94093e-27
GTD_{s1}	1-9.96838e-1	1-5.00000e-1	1-7.42619e-6	1-7.42619e-6	1-7.45394e-3	1-4.92670e-1	1-2.97046e-27
GTD_{s2}	1-2.12176e-6	1-7.42619e-6	1-7.42616e-12	1-7.42616e-8	1-7.42619e-6	1-7.42644e-5	1-7.42616e-28
GTD_{d1}	1-9.96948e-1	1-5.00001e-1	1-7.42619e-6	1-7.42619e-6	1-7.45395e-3	1-4.92670e-1	1-2.97047e-27
GTD_{d2}	1-2.12176e-6	1-7.42619e-6	1-7.42617e-12	1-7.42617e-8	1-7.42619e-6	1-7.42644e-5	1-7.42617e-28
DTD_1	9.99995e-1	8.66026e-1	3.85387e-3	3.85387e-3	1.21870e-1	8.61752e-1	7.70774e-14
DTD_2	2.05998e-3	3.85387e-3	3.85387e-6	3.85387e-4	3.85387e-3	1.21870e-2	3.85387e-14

Table 2 Derived ratios across cases.

Ratio	Case 1 $M=M$, $\sim r_s$	Case 2 $M=M$, $\sim \frac{4}{3} r_s$	Case 3 $R=R$	Case 4 $\rho=\rho$	Case 5 $I=I$	Case 6 General	Case 7 Classical
$\mathrm{DTD}_1/\mathrm{DTD}_2$	4.85439e+2	2.24716e+2	1.00000e+3	1.00000e+1	3.16228e+1	7.07107e+1	2.00000e+0
$\sqrt{D_1/D_2}$	4.85439e+2	2.24716e+2	1.00000e+3	1.00000e+1	3.16228e+1	7.07107e+1	2.00000e+0
$\sqrt{(M_1R_2)/(M_2R_1)}$	4.85439e+2	2.24716e+2	1.00000e+3	1.00000e+1	3.16228e+1	7.07107e+1	2.00000e+0
$\sqrt{(I_1/I_2)(R_2/R_1)^3}$	4.85439e+2	2.24716e+2	1.00000e+3	1.00000e+1	3.16228e+1	7.07107e+1	2.00000e+0
$(I_1/I_2)^{1/5}(\rho_1/\rho_2)^{3/10}$	4.85439e+2	2.24716e+2	1.00000e+3	1.00000e+1	3.16228e+1	7.07107e+1	2.00000e+0
$(M_1R_2)/(M_2R_1)$	2.35651e+5	5.04972e+4	1.00000e+6	1.00000e+2	1.00000e+3	5.00000e+3	4.00000e+0
D_1/D_2	2.35651e+5	5.04972e+4	1.00000e+6	1.00000e+2	1.00000e+3	5.00000e+3	4.00000e+0
$(I_1/I_2)(R_2/R_1)^3$	2.35651e+5	5.04972e+4	1.00000e+6	1.00000e+2	1.00000e+3	5.00000e+3	4.00000e+0
$\mathrm{GTD}_{s1}/\mathrm{GTD}_{s2}$	1-9.96838e-1	1-4.99996e-1	1-7.42618e-6	1-7.35193e-6	1-7.44657e-3	1-4.92632e-1	1-2.22785e-27
$\mathrm{GTD}_{d1}/\mathrm{GTD}_{d2}$	1-9.96948e-1	1-4.99997e-1	1-7.42619e-6	1-7.35193e-6	1-7.44658e-3	1-4.92633e-1	1-2.22785e-27
$\sqrt{1-\mathrm{DTD}_1^2}/\sqrt{1-\mathrm{DTD}_2^2}$	1-9.96948e-1	1-4.99997e-1	1-7.42619e-6	1-7.35193e-6	1-7.44658e-3	1-4.92633e-1	1-2.22785e-27
I_1/I_2	1.80078e-11	3.92163e-10	1.00000e+6	1.00000e+5	1.00000e+0	4.00000e-38	3.20000e+1
$(I_1/I_2)^{1/2}$	4.24356e-6	1.98031e-5	1.00000e+3	3.16228e+2	1.00000e+0	2.00000e-19	5.65685e+0
$(\rho_1/\rho_2)^{1/5}(R_1/R_2)$	7.09728e-3	1.31430e-2	1.58489e+1	1.00000e+1	1.00000e+0	3.31445e-8	2.00000e+0
$(I_1/I_2)^{1/5}$	7.09728e-3	1.31430e-2	1.58489e+1	1.00000e+1	1.00000e+0	3.31445e-8	2.00000e+0
M_1/M_2	1.00000e+0	1.00000e+0	1.00000e+6	1.00000e+3	1.00000e+2	1.00000e-10	8.00000e+0
$\sqrt{M_1/M_2}$	1.00000e+0	1.00000e+0	1.00000e+3	3.16228e+1	1.00000e+1	1.00000e-5	2.82843e+0
R_1/R_2	4.24356e-6	1.98031e-5	1.00000e+0	1.00000e+1	1.00000e-1	2.00000e-14	2.00000e+0
$\sqrt{R_1/R_2}$	2.05999e-3	4.45007e-3	1.00000e+0	3.16228e+0	3.16228e-1	1.41421e-7	1.41421e+0

5.1 Case 1: ($M_1 = M_2$, $R_1 \approx r_s$)

The two identical masses cancel, reducing the DeGerlia inertial density ratio down to a ratio of radii.

$$k_D = \frac{M_1 R_2}{M_2 R_1 D_{crit}} = \frac{R_2}{R_1 D_{crit}}$$

Note, this reflects the k calculation for gravitational time dilation, where we are comparing with r_s. Because $GTD = 0$ exactly at r_s, we set the comparison radius to $1.00001\, r_s$, an "arbitrarily" small amount above r_s.

System properties:

	System 1 (S_1)	System 2 (S_2)
M (kg)	2.00000e30	2.00000e30
R (m)	2.97049e3	7.00000e8

Calculate DTD for each System:

$$DTD = \sqrt{k} \quad \text{where} \quad k = M/(RD_{crit})$$
$$k_1 = \frac{2.00000\text{e}30\,\text{kg}}{(2.97049\text{e}3\,\text{m})(6.73295\text{e}26\,\text{kg/m})} = 9.9999068441\text{e}{-1}$$
$$DTD_1 = \sqrt{9.9999068441\text{e}{-1}} = 9.9999534220\text{e}{-1}$$
$$k_2 = \frac{2.00000\text{e}30\,\text{kg}}{(7.00000\text{e}8\,\text{m})(6.73295\text{e}26\,\text{kg/m})} = 4.24352\text{e}{-6}$$
$$DTD_2 = \sqrt{4.24352\text{e}{-6}} = 2.05998\text{e}{-3}$$
$$\frac{DTD_1}{DTD_2} = \frac{9.9999534220\text{e}{-1}}{2.05998\text{e}{-3}} = 4.85439\text{e}2$$

Calculate GTD ratio from DTD:

$$\frac{GTD_{d1}}{GTD_{d2}} = \frac{\sqrt{1-DTD_1^2}}{\sqrt{1-DTD_2^2}} = \frac{\sqrt{1-(9.9999534220\text{e}{-1})^2}}{\sqrt{1-(2.05998\text{e}{-3})^2}}$$
$$= \frac{\sqrt{1-9.9999068441\text{e}{-1}}}{\sqrt{1-4.24352\text{e}{-6}}}$$
$$= \frac{\sqrt{9.31559\text{e}{-6}}}{\sqrt{9.9999575648\text{e}{-1}}}$$
$$= \frac{3.05214\text{e}{-3}}{9.9999787824\text{e}{-1}} = 3.05215\text{e}{-3}$$
$$= 1 - 9.9694785\text{e}{-1}$$

Verify by Calculating GTD from Schwarzschild Radius Ratio:

Schwarzschild radii:

$$r_{s1} = r_{s2} = \frac{2GM}{c^2} = \frac{2(6.67430\text{e}{-11}\,\text{m}^3\,\text{kg}^{-1}\,\text{s}^{-2})(2.00000\text{e}30\,\text{kg})}{(2.99792\text{e}8\,\text{m}\,\text{s}^{-1})^2}$$
$$= 2.97046\text{e}3\,\text{m}$$

Calculate k values from the r_s/R ratio:

$$k_1 = \frac{r_{s1}}{R_1} = \frac{2.97046\text{e}3\,\text{m}}{2.97049\text{e}3\,\text{m}} = 9.9999000010\text{e}{-1}$$
$$k_2 = \frac{r_{s2}}{R_2} = \frac{2.97046\text{e}3\,\text{m}}{7.00000\text{e}8\,\text{m}} = 4.24352\text{e}{-6}$$

Calculate GTD via Schwarzschild radius ratio:

$$GTD_{s1} = \sqrt{1-k_1} = \sqrt{9.99990\text{e}{-6}} = 3.16226\text{e}{-3} = 1 - 9.9683774\text{e}{-1}$$
$$GTD_{s2} = \sqrt{1-k_2} = \sqrt{9.9999575648\text{e}{-1}} = 9.9999787824\text{e}{-1}$$
$$= 1 - 2.12176\text{e}{-6}$$
$$\frac{GTD_{s1}}{GTD_{s2}} = \frac{3.16226\text{e}{-3}}{9.9999787824\text{e}{-1}} = 3.16227\text{e}{-3}$$
$$= 1 - 9.9683773\text{e}{-1}$$

Time-dilation ratio between System 1 and System 2, computed two ways:

$$\frac{GTD_{s1}}{GTD_{s2}} = 1 - 9.9683773\text{e}{-1} \quad (\text{via } r_s)$$
$$\frac{GTD_{d1}}{GTD_{d2}} = 1 - 9.9694785\text{e}{-1} \quad (\text{via } D_{crit})$$

5.2 Case 2: ($M_1 = M_2$, $R_1 = \frac{4}{3} r_s$)

The two identical masses cancel, reducing the DeGerlia inertial density ratio down to a ratio of radii.

$$k_D = \frac{M_1 R_2}{M_2 R_1 D_{crit}} = \frac{R_2}{R_1 D_{crit}}$$

Note, This reflects the k calculation for gravitational time dilation. However, in this case, instead of comparing with r_s, we are comparing with $\frac{4}{3} r_s$, which equates to a gravitational time dilation of $GTD = 5.00000\text{e}{-1}$.

System properties:

	System 1 (S_1)	System 2 (S_2)
M (kg)	1.00000e30	1.00000e30
R (m)	1.98031e3	1.00000e8

Calculate DTD for each System:

$$DTD = \sqrt{k} \quad \text{where} \quad k = M/(RD_{crit})$$
$$k_1 = \frac{1.00000\text{e}30\,\text{kg}}{(1.98031\text{e}3\,\text{m})(6.73295\text{e}26\,\text{kg/m})} = 7.50001\text{e}{-1}$$
$$DTD_1 = \sqrt{7.50001\text{e}{-1}} = 8.66026\text{e}{-1}$$
$$k_2 = \frac{1.00000\text{e}30\,\text{kg}}{(1.00000\text{e}8\,\text{m})(6.73295\text{e}26\,\text{kg/m})} = 1.48523\text{e}{-5}$$
$$DTD_2 = \sqrt{1.48523\text{e}{-5}} = 3.85387\text{e}{-3}$$
$$\frac{DTD_1}{DTD_2} = \frac{8.66026\text{e}{-1}}{3.85387\text{e}{-3}} = 2.24716\text{e}2$$

Calculate GTD ratio from DTD:

$$\frac{GTD_{d1}}{GTD_{d2}} = \frac{\sqrt{1-DTD_1^2}}{\sqrt{1-DTD_2^2}} = \frac{\sqrt{1-(8.66026\text{e}{-1})^2}}{\sqrt{1-(3.85387\text{e}{-3})^2}}$$
$$= \frac{\sqrt{1-7.50001\text{e}{-1}}}{\sqrt{1-1.48523\text{e}{-5}}}$$
$$= \frac{\sqrt{2.49999\text{e}{-1}}}{\sqrt{9.999851477\text{e}{-1}}}$$
$$= \frac{4.99999\text{e}{-1}}{9.9999257381\text{e}{-1}} = 5.00003\text{e}{-1}$$
$$= 1 - 4.99997\text{e}{-1}$$

Verify by Calculating GTD from Schwarzschild Radius Ratio:

Schwarzschild radii:

$$r_{s1} = r_{s2} = \frac{2GM}{c^2} = \frac{2(6.67430\text{e}{-11}\,\text{m}^3\,\text{kg}^{-1}\,\text{s}^{-2})(1.00000\text{e}30\,\text{kg})}{(2.99792\text{e}8\,\text{m}\,\text{s}^{-1})^2}$$
$$= 1.48523\text{e}3\,\text{m}$$

Calculate k values from the r_s/R ratio:

$$k_1 = \frac{r_{s1}}{R_1} = \frac{1.48523\text{e}3\,\text{m}}{1.98031\text{e}3\,\text{m}} = 7.50000\text{e}{-1}$$
$$k_2 = \frac{r_{s2}}{R_2} = \frac{1.48523\text{e}3\,\text{m}}{1.00000\text{e}8\,\text{m}} = 1.48523\text{e}{-5}$$

Calculate GTD via Schwarzschild radius ratio:

$$GTD_{s1} = \sqrt{1-k_1} = \sqrt{2.50000\text{e}{-1}} = 5.00000\text{e}{-1} = 1 - 5.00000\text{e}{-1}$$
$$GTD_{s2} = \sqrt{1-k_2} = \sqrt{9.999851477\text{e}{-1}} = 9.9999257381\text{e}{-1}$$
$$= 1 - 7.42619\text{e}{-6}$$
$$\frac{GTD_{s1}}{GTD_{s2}} = \frac{5.00000\text{e}{-1}}{9.9999257381\text{e}{-1}} = 5.00004\text{e}{-1}$$
$$= 1 - 4.99996\text{e}{-1}$$

Time-dilation ratio between System 1 and System 2, computed two ways:

$$\frac{GTD_{s1}}{GTD_{s2}} = 1 - 4.99996\text{e}{-1} \quad (\text{via } r_s)$$
$$\frac{GTD_{d1}}{GTD_{d2}} = 1 - 4.99997\text{e}{-1} \quad (\text{via } D_{crit})$$

5.3 Case 3: ($R_1 = R_2$)

When the two systems share the same radius, the radius terms cancel in k_D:

$$k_D = \frac{M_1 R_2}{M_2 R_1 D_{crit}} = \frac{M_1}{M_2 D_{crit}}$$

The Lorentz factor has the same form as Equation 17, with $k = v^2/c^2$. Since velocity is distance per unit time, v^2/c^2 is the square of a ratio of two lengths, with the shared time unit cancelling.

System properties:

	System 1 (S_1)	System 2 (S_2)
M (kg)	1.00000e30	1.00000e24
R (m)	1.00000e8	1.00000e8

Calculate DTD for each System:

$$\begin{aligned}
DTD &= \sqrt{k} \quad \text{where} \quad k = M/(RD_{crit}) \\
k_1 &= \frac{1.00000\text{e}30\,\text{kg}}{(1.00000\text{e}8\,\text{m})(6.73295\text{e}26\,\text{kg/m})} = 1.48523\text{e}{-5} \\
DTD_1 &= \sqrt{1.48523\text{e}{-5}} = 3.85387\text{e}{-3} \\
k_2 &= \frac{1.00000\text{e}24\,\text{kg}}{(1.00000\text{e}8\,\text{m})(6.73295\text{e}26\,\text{kg/m})} = 1.48523\text{e}{-11} \\
DTD_2 &= \sqrt{1.48523\text{e}{-11}} = 3.85387\text{e}{-6} \\
\frac{DTD_1}{DTD_2} &= \frac{3.85387\text{e}{-3}}{3.85387\text{e}{-6}} = 1.00000\text{e}3
\end{aligned}$$

Calculate GTD ratio from DTD:

$$\begin{aligned}
\frac{GTD_{d1}}{GTD_{d2}} &= \frac{\sqrt{1-DTD_1^2}}{\sqrt{1-DTD_2^2}} = \frac{\sqrt{1-(3.85387\text{e}{-3})^2}}{\sqrt{1-(3.85387\text{e}{-6})^2}} \\
&= \frac{\sqrt{1-1.48523\text{e}{-5}}}{\sqrt{1-1.48523\text{e}{-11}}} \\
&= \frac{\sqrt{9.999851477\text{e}{-1}}}{\sqrt{9.999999999851477\text{e}{-1}}} \\
&= \frac{9.9999257381\text{e}{-1}}{9.9999999999257383\text{e}{-1}} = 9.9999257381\text{e}{-1} \\
&= 1 - 7.42619\text{e}{-6}
\end{aligned}$$

Verify by Calculating GTD from Schwarzschild Radius Ratio:

Schwarzschild radii:

$$\begin{aligned}
r_{s1} &= \frac{2GM_1}{c^2} = \frac{2(6.67430\text{e}{-11}\,\text{m}^3\,\text{kg}^{-1}\,\text{s}^{-2})(1.00000\text{e}30\,\text{kg})}{(2.99792\text{e}8\,\text{m}\,\text{s}^{-1})^2} \\
&= 1.48523\text{e}3\,\text{m} \\
r_{s2} &= \frac{2GM_2}{c^2} = \frac{2(6.67430\text{e}{-11}\,\text{m}^3\,\text{kg}^{-1}\,\text{s}^{-2})(1.00000\text{e}24\,\text{kg})}{(2.99792\text{e}8\,\text{m}\,\text{s}^{-1})^2} \\
&= 1.48523\text{e}{-3}\,\text{m}
\end{aligned}$$

Calculate k values from the r_s/R ratio:

$$\begin{aligned}
k_1 &= \frac{r_{s1}}{R_1} &&= \frac{1.48523\text{e}3\,\text{m}}{1.00000\text{e}8\,\text{m}} &&= 1.48523\text{e}{-5} \\
k_2 &= \frac{r_{s2}}{R_2} &&= \frac{1.48523\text{e}{-3}\,\text{m}}{1.00000\text{e}8\,\text{m}} &&= 1.48523\text{e}{-11}
\end{aligned}$$

Calculate GTD via Schwarzschild radius ratio:

$$\begin{aligned}
GTD_{s1} &= \sqrt{1-k_1} = \sqrt{9.999851477\text{e}{-1}} = 9.9999257381\text{e}{-1} \\
&= 1 - 7.42619\text{e}{-6} \\
GTD_{s2} &= \sqrt{1-k_2} = \sqrt{9.999999999851477\text{e}{-1}} \\
&= 9.9999999999257384\text{e}{-1} = 1 - 7.42616\text{e}{-12} \\
\frac{GTD_{s1}}{GTD_{s2}} &= \frac{9.9999257381\text{e}{-1}}{9.9999999999257384\text{e}{-1}} \\
&= 9.9999257382\text{e}{-1} = 1 - 7.42618\text{e}{-6}
\end{aligned}$$

Time-dilation ratio between System 1 and System 2, computed two ways:

$$\begin{aligned}
\frac{GTD_{s1}}{GTD_{s2}} &= 1 - 7.42618\text{e}{-6} \quad (\text{via } r_s) \\
\frac{GTD_{d1}}{GTD_{d2}} &= 1 - 7.42619\text{e}{-6} \quad (\text{via } D_{crit})
\end{aligned}$$

5.4 Case 4: ($\rho_1 = \rho_2$)

When the two systems share the same density, $M \propto R^3$, so $M_1/M_2 = (R_1/R_2)^3$. Substituting into k_D:

$$k_D = \frac{M_1 R_2}{M_2 R_1 D_{crit}} = \left(\frac{R_1}{R_2}\right)^3 \cdot \frac{R_2}{R_1 D_{crit}} = \frac{1}{D_{crit}} \left(\frac{R_1}{R_2}\right)^2$$

The DeGerlia inertial density ratio reduces to the square of the radius ratio; the linear scale factor follows as $\sqrt{k_D} = R_1/R_2$, recovering the isometric equivalence.

System properties:

	System 1 (S_1)	System 2 (S_2)
M (kg)	1.00000e30	1.00000e27
R (m)	1.00000e8	1.00000e7

Calculate DTD for each System:

$$\begin{aligned}
DTD &= \sqrt{k} \quad \text{where} \quad k = M/(RD_{crit}) \\
k_1 &= \frac{1.00000\text{e}30\,\text{kg}}{(1.00000\text{e}8\,\text{m})(6.73295\text{e}26\,\text{kg/m})} = 1.48523\text{e}{-5} \\
DTD_1 &= \sqrt{1.48523\text{e}{-5}} = 3.85387\text{e}{-3} \\
k_2 &= \frac{1.00000\text{e}27\,\text{kg}}{(1.00000\text{e}7\,\text{m})(6.73295\text{e}26\,\text{kg/m})} = 1.48523\text{e}{-7} \\
DTD_2 &= \sqrt{1.48523\text{e}{-7}} = 3.85387\text{e}{-4} \\
\frac{DTD_1}{DTD_2} &= \frac{3.85387\text{e}{-3}}{3.85387\text{e}{-4}} = 1.00000\text{e}1
\end{aligned}$$

Calculate GTD ratio from DTD:

$$\begin{aligned}
\frac{GTD_{d1}}{GTD_{d2}} &= \frac{\sqrt{1-DTD_1^2}}{\sqrt{1-DTD_2^2}} = \frac{\sqrt{1-(3.85387\text{e}{-3})^2}}{\sqrt{1-(3.85387\text{e}{-4})^2}} \\
&= \frac{\sqrt{1-1.48523\text{e}{-5}}}{\sqrt{1-1.48523\text{e}{-7}}} \\
&= \frac{\sqrt{9.999851477\text{e}{-1}}}{\sqrt{9.99999851477\text{e}{-1}}} \\
&= \frac{9.9999257381\text{e}{-1}}{9.999999257383\text{e}{-1}} = 9.9999264807\text{e}{-1} \\
&= 1 - 7.35193\text{e}{-6}
\end{aligned}$$

Verify by Calculating GTD from Schwarzschild Radius Ratio:

Schwarzschild radii:

$$\begin{aligned}
r_{s1} &= \frac{2GM_1}{c^2} = \frac{2(6.67430\text{e}{-11}\,\text{m}^3\,\text{kg}^{-1}\,\text{s}^{-2})(1.00000\text{e}30\,\text{kg})}{(2.99792\text{e}8\,\text{m}\,\text{s}^{-1})^2} \\
&= 1.48523\text{e}3\,\text{m} \\
r_{s2} &= \frac{2GM_2}{c^2} = \frac{2(6.67430\text{e}{-11}\,\text{m}^3\,\text{kg}^{-1}\,\text{s}^{-2})(1.00000\text{e}27\,\text{kg})}{(2.99792\text{e}8\,\text{m}\,\text{s}^{-1})^2} \\
&= 1.48523\,\text{m}
\end{aligned}$$

Calculate k values from the r_s/R ratio:

$$\begin{aligned}
k_1 &= \frac{r_{s1}}{R_1} &&= \frac{1.48523\text{e}3\,\text{m}}{1.00000\text{e}8\,\text{m}} &&= 1.48523\text{e}{-5} \\
k_2 &= \frac{r_{s2}}{R_2} &&= \frac{1.48523\,\text{m}}{1.00000\text{e}7\,\text{m}} &&= 1.48523\text{e}{-7}
\end{aligned}$$

Calculate GTD via Schwarzschild radius ratio:

$$\begin{aligned}
GTD_{s1} &= \sqrt{1-k_1} = \sqrt{9.999851477\text{e}{-1}} = 9.9999257381\text{e}{-1} \\
&= 1 - 7.42619\text{e}{-6} \\
GTD_{s2} &= \sqrt{1-k_2} = \sqrt{9.99999851477\text{e}{-1}} = 9.999999257384\text{e}{-1} \\
&= 1 - 7.42616\text{e}{-8} \\
\frac{GTD_{s1}}{GTD_{s2}} &= \frac{9.9999257381\text{e}{-1}}{9.999999257384\text{e}{-1}} \\
&= 9.9999264807\text{e}{-1} = 1 - 7.35193\text{e}{-6}
\end{aligned}$$

Time-dilation ratio between System 1 and System 2, computed two ways:

$$\begin{aligned}
\frac{GTD_{s1}}{GTD_{s2}} &= 1 - 7.35193\text{e}{-6} \quad (\text{via } r_s) \\
\frac{GTD_{d1}}{GTD_{d2}} &= 1 - 7.35193\text{e}{-6} \quad (\text{via } D_{crit})
\end{aligned}$$

5.5 Case 5: ($I_1 = I_2$)

When the two systems share the same moment of inertia, $M_1 R_1^2 = M_2 R_2^2$, so $M_1/M_2 = (R_2/R_1)^2$. Substituting into k_D:

$$k_D = \frac{M_1 R_2}{M_2 R_1 D_{crit}} = \left(\frac{R_2}{R_1}\right)^2 \cdot \frac{R_2}{R_1 D_{crit}} = \frac{1}{D_{crit}} \left(\frac{R_2}{R_1}\right)^3$$

The DeGerlia inertial density ratio reduces to the cube of the radius ratio.

System properties:

	System 1 (S_1)	System 2 (S_2)
M (kg)	1.00000e32	1.00000e30
R (m)	1.00000e7	1.00000e8

Calculate DTD for each System:

$$DTD = \sqrt{k} \quad \text{where} \quad k = M/(RD_{crit})$$
$$k_1 = \frac{1.00000\text{e}32\,\text{kg}}{(1.00000\text{e}7\,\text{m})(6.73295\text{e}26\,\text{kg/m})} = 1.48523\text{e}{-2}$$
$$DTD_1 = \sqrt{1.48523\text{e}{-2}} = 1.21870\text{e}{-1}$$
$$k_2 = \frac{1.00000\text{e}30\,\text{kg}}{(1.00000\text{e}8\,\text{m})(6.73295\text{e}26\,\text{kg/m})} = 1.48523\text{e}{-5}$$
$$DTD_2 = \sqrt{1.48523\text{e}{-5}} = 3.85387\text{e}{-3}$$
$$\frac{DTD_1}{DTD_2} = \frac{1.21870\text{e}{-1}}{3.85387\text{e}{-3}} = 3.16228\text{e}1$$

Calculate GTD ratio from DTD:

$$\begin{aligned}
\frac{GTD_{d1}}{GTD_{d2}} &= \frac{\sqrt{1 - DTD_1^2}}{\sqrt{1 - DTD_2^2}} = \frac{\sqrt{1 - (1.21870\text{e}{-1})^2}}{\sqrt{1 - (3.85387\text{e}{-3})^2}} \\
&= \frac{\sqrt{1 - 1.48523\text{e}{-2}}}{\sqrt{1 - 1.48523\text{e}{-5}}} \\
&= \frac{\sqrt{9.851477\text{e}{-1}}}{\sqrt{9.999851477\text{e}{-1}}} \\
&= \frac{9.9254605\text{e}{-1}}{9.9999257381\text{e}{-1}} = 9.9255342\text{e}{-1} \\
&= 1 - 7.44658\text{e}{-3}
\end{aligned}$$

Verify by Calculating GTD from Schwarzschild Radius Ratio:

Schwarzschild radii:

$$r_{s1} = \frac{2GM_1}{c^2} = \frac{2(6.67430\text{e}{-11}\,\text{m}^3\,\text{kg}^{-1}\,\text{s}^{-2})(1.00000\text{e}32\,\text{kg})}{(2.99792\text{e}8\,\text{m}\,\text{s}^{-1})^2}$$
$$= 1.48523\text{e}5\,\text{m}$$
$$r_{s2} = \frac{2GM_2}{c^2} = \frac{2(6.67430\text{e}{-11}\,\text{m}^3\,\text{kg}^{-1}\,\text{s}^{-2})(1.00000\text{e}30\,\text{kg})}{(2.99792\text{e}8\,\text{m}\,\text{s}^{-1})^2}$$
$$= 1.48523\text{e}3\,\text{m}$$

Calculate k values from the r_s/R ratio:

$$k_1 = \frac{r_{s1}}{R_1} = \frac{1.48523\text{e}5\,\text{m}}{1.00000\text{e}7\,\text{m}} = 1.48523\text{e}{-2}$$
$$k_2 = \frac{r_{s2}}{R_2} = \frac{1.48523\text{e}3\,\text{m}}{1.00000\text{e}8\,\text{m}} = 1.48523\text{e}{-5}$$

Calculate GTD via Schwarzschild radius ratio:

$$GTD_{s1} = \sqrt{1 - k_1} = \sqrt{9.851477\text{e}{-1}} = 9.9254606\text{e}{-1} = 1 - 7.45394\text{e}{-3}$$
$$GTD_{s2} = \sqrt{1 - k_2} = \sqrt{9.999851477\text{e}{-1}} = 9.9999257381\text{e}{-1}$$
$$= 1 - 7.42619\text{e}{-6}$$
$$\frac{GTD_{s1}}{GTD_{s2}} = \frac{9.9254606\text{e}{-1}}{9.9999257381\text{e}{-1}} = 9.9255343\text{e}{-1}$$
$$= 1 - 7.44657\text{e}{-3}$$

Time-dilation ratio between System 1 and System 2, computed two ways:

$$\frac{GTD_{s1}}{GTD_{s2}} = 1 - 7.44657\text{e}{-3} \quad (\text{via } r_s)$$
$$\frac{GTD_{d1}}{GTD_{d2}} = 1 - 7.44658\text{e}{-3} \quad (\text{via } D_{crit})$$

5.6 Case 6: (General, unconstrained)

When no constraint is applied between the systems, k_D stays in its full form:

$$k_D = \frac{M_1 R_2}{M_2 R_1 D_{crit}}$$

Both mass and radius must be specified independently for each system.

System properties:

	System 1 (S_1)	System 2 (S_2)
M (kg)	1.00000e20	1.00000e30
R (m)	2.00000e−7	1.00000e7

Calculate DTD for each System:

$$DTD = \sqrt{k} \quad \text{where} \quad k = M/(RD_{crit})$$
$$k_1 = \frac{1.00000\text{e}20\,\text{kg}}{(2.00000\text{e}{-7}\,\text{m})(6.73295\text{e}26\,\text{kg/m})} = 7.42617\text{e}{-1}$$
$$DTD_1 = \sqrt{7.42617\text{e}{-1}} = 8.61752\text{e}{-1}$$
$$k_2 = \frac{1.00000\text{e}30\,\text{kg}}{(1.00000\text{e}7\,\text{m})(6.73295\text{e}26\,\text{kg/m})} = 1.48523\text{e}{-4}$$
$$DTD_2 = \sqrt{1.48523\text{e}{-4}} = 1.21870\text{e}{-2}$$
$$\frac{DTD_1}{DTD_2} = \frac{8.61752\text{e}{-1}}{1.21870\text{e}{-2}} = 7.07107\text{e}1$$

Calculate GTD ratio from DTD:

$$\begin{aligned}
\frac{GTD_{d1}}{GTD_{d2}} &= \frac{\sqrt{1 - DTD_1^2}}{\sqrt{1 - DTD_2^2}} = \frac{\sqrt{1 - (8.61752\text{e}{-1})^2}}{\sqrt{1 - (1.21870\text{e}{-2})^2}} \\
&= \frac{\sqrt{1 - 7.42617\text{e}{-1}}}{\sqrt{1 - 1.48523\text{e}{-4}}} \\
&= \frac{\sqrt{2.57383\text{e}{-1}}}{\sqrt{9.99851477\text{e}{-1}}} \\
&= \frac{5.07330\text{e}{-1}}{9.999257356\text{e}{-1}} = 5.07367\text{e}{-1} \\
&= 1 - 4.92633\text{e}{-1}
\end{aligned}$$

Verify by Calculating GTD from Schwarzschild Radius Ratio:

Schwarzschild radii:

$$r_{s1} = \frac{2GM_1}{c^2} = \frac{2(6.67430\text{e}{-11}\,\text{m}^3\,\text{kg}^{-1}\,\text{s}^{-2})(1.00000\text{e}20\,\text{kg})}{(2.99792\text{e}8\,\text{m}\,\text{s}^{-1})^2}$$
$$= 1.48523\text{e}{-7}\,\text{m}$$
$$r_{s2} = \frac{2GM_2}{c^2} = \frac{2(6.67430\text{e}{-11}\,\text{m}^3\,\text{kg}^{-1}\,\text{s}^{-2})(1.00000\text{e}30\,\text{kg})}{(2.99792\text{e}8\,\text{m}\,\text{s}^{-1})^2}$$
$$= 1.48523\text{e}3\,\text{m}$$

Calculate k values from the r_s/R ratio:

$$k_1 = \frac{r_{s1}}{R_1} = \frac{1.48523\text{e}{-7}\,\text{m}}{2.00000\text{e}{-7}\,\text{m}} = 7.42616\text{e}{-1}$$
$$k_2 = \frac{r_{s2}}{R_2} = \frac{1.48523\text{e}3\,\text{m}}{1.00000\text{e}7\,\text{m}} = 1.48523\text{e}{-4}$$

Calculate GTD via Schwarzschild radius ratio:

$$GTD_{s1} = \sqrt{1 - k_1} = \sqrt{2.57384\text{e}{-1}} = 5.07330\text{e}{-1} = 1 - 4.92670\text{e}{-1}$$
$$GTD_{s2} = \sqrt{1 - k_2} = \sqrt{9.99851477\text{e}{-1}} = 9.999257356\text{e}{-1}$$
$$= 1 - 7.42644\text{e}{-5}$$
$$\frac{GTD_{s1}}{GTD_{s2}} = \frac{5.07330\text{e}{-1}}{9.999257356\text{e}{-1}} = 5.07368\text{e}{-1}$$
$$= 1 - 4.92632\text{e}{-1}$$

Time-dilation ratio between System 1 and System 2, computed two ways:

$$\frac{GTD_{s1}}{GTD_{s2}} = 1 - 4.92632\text{e}{-1} \quad (\text{via } r_s)$$
$$\frac{GTD_{d1}}{GTD_{d2}} = 1 - 4.92633\text{e}{-1} \quad (\text{via } D_{crit})$$

5.7 Case 7: (Classical)

System properties:

	System 1 (S_1)	System 2 (S_2)
M (kg)	8.00000	1.00000
R (m)	2.00000	1.00000

Calculate DTD for each System:

$$DTD = \sqrt{k} \quad \text{where} \quad k = M/(RD_{crit})$$
$$k_1 = \frac{8.00000\,\text{kg}}{(2.00000\,\text{m})(6.73295\text{e}26\,\text{kg/m})} = 5.94093\text{e}{-27}$$
$$DTD_1 = \sqrt{5.94093\text{e}{-27}} = 7.70774\text{e}{-14}$$
$$k_2 = \frac{1.00000\,\text{kg}}{(1.00000\,\text{m})(6.73295\text{e}26\,\text{kg/m})} = 1.48523\text{e}{-27}$$
$$DTD_2 = \sqrt{1.48523\text{e}{-27}} = 3.85387\text{e}{-14}$$
$$\frac{DTD_1}{DTD_2} = \frac{7.70774\text{e}{-14}}{3.85387\text{e}{-14}} = 2.00000$$

Calculate GTD ratio from DTD:

$$\frac{GTD_{d1}}{GTD_{d2}} = \frac{\sqrt{1-DTD_1^2}}{\sqrt{1-DTD_2^2}} = \frac{\sqrt{1-(7.70774\text{e}{-14})^2}}{\sqrt{1-(3.85387\text{e}{-14})^2}}$$
$$= \frac{\sqrt{1-5.94093\text{e}{-27}}}{\sqrt{1-1.48523\text{e}{-27}}}$$
$$= \frac{\sqrt{9.99999999999999999999999999940590 7\text{e}{-1}}}{\sqrt{9.999999999999999999999999999985147 7\text{e}{-1}}}$$
$$= \frac{9.999999999999999999999999999970295 3\text{e}{-1}}{9.9999999999999999999999999999925738 3\text{e}{-1}}$$
$$= 9.999999999999999999999999999977721 5\text{e}{-1}$$
$$= 1 - 2.22785\text{e}{-27}$$

Verify by Calculating GTD from Schwarzschild Radius Ratio:

Schwarzschild radii:

$$r_{s1} = \frac{2GM_1}{c^2} = \frac{2(6.67430\text{e}{-11}\,\text{m}^3\,\text{kg}^{-1}\,\text{s}^{-2})(8.00000\,\text{kg})}{(2.99792\text{e}8\,\text{m}\,\text{s}^{-1})^2}$$
$$= 1.18819\text{e}{-26}\,\text{m}$$
$$r_{s2} = \frac{2GM_2}{c^2} = \frac{2(6.67430\text{e}{-11}\,\text{m}^3\,\text{kg}^{-1}\,\text{s}^{-2})(1.00000\,\text{kg})}{(2.99792\text{e}8\,\text{m}\,\text{s}^{-1})^2}$$
$$= 1.48523\text{e}{-27}\,\text{m}$$

Calculate k values from the r_s/R ratio:

$$k_1 = \frac{r_{s1}}{R_1} = \frac{1.18819\text{e}{-26}\,\text{m}}{2.00000\,\text{m}} = 5.94093\text{e}{-27}$$
$$k_2 = \frac{r_{s2}}{R_2} = \frac{1.48523\text{e}{-27}\,\text{m}}{1.00000\,\text{m}} = 1.48523\text{e}{-27}$$

Calculate GTD via Schwarzschild radius ratio:

$$GTD_{s1} = \sqrt{1-k_1}$$
$$= \sqrt{9.99999999999999999999999999940590 7\text{e}{-1}}$$
$$= 9.999999999999999999999999999970295 4\text{e}{-1}$$
$$= 1 - 2.97046\text{e}{-27}$$
$$GTD_{s2} = \sqrt{1-k_2}$$
$$= \sqrt{9.999999999999999999999999999985147 7\text{e}{-1}}$$
$$= 9.9999999999999999999999999999925738 4\text{e}{-1}$$
$$= 1 - 7.42616\text{e}{-28}$$
$$\frac{GTD_{s1}}{GTD_{s2}} = \frac{9.999999999999999999999999999970295 4\text{e}{-1}}{9.9999999999999999999999999999925738 4\text{e}{-1}}$$
$$= 9.999999999999999999999999999977721 5\text{e}{-1}$$
$$= 1 - 2.22785\text{e}{-27}$$

Time-dilation ratio between System 1 and System 2, computed two ways:

$$\frac{GTD_{s1}}{GTD_{s2}} = 1 - 2.22785\text{e}{-27} \quad (\text{via } r_s)$$
$$\frac{GTD_{d1}}{GTD_{d2}} = 1 - 2.22785\text{e}{-27} \quad (\text{via } D_{crit})$$

6 Analysis

The reformulation is meant to carry three points, and nothing more ambitious. First, it is a pace-of-time framework that is simply easier to work with: expressed in DTD and D, comparing two systems is a direct ratio of geometries, with no benchmark, no subtraction-from-one, and — because D_{crit} cancels — no dependence on G. One returns to GTD, the $\sqrt{1-k}$ form, only at the final step, to convert relative pace into relative clock time. Second, general relativity is always two-system; we merely masquerade it as one. Conventional gravitational time dilation looks like a single-system calculation only because it silently fixes the second system — the same mass at its Schwarzschild radius, where $DTD = 1$. Make that benchmark explicit and the two-system structure, present all along, is plain. Third, treating it as one system is a disservice to the broader mathematics: the collapse to a single system rests on presuming that some property of the hidden second system is a fixed, universal given, a presumption that obscures the real structure and perpetuates a misconception that dilutes the core principle. The honest formulation keeps both systems in view and recognizes the single-system shortcut as a convenience, not as the underlying physics.

7 Conclusions

8 Hypotheses of Explanation

9 Further Exploration

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Conflicts of Interest

The author declares no competing interests.

Data Availability

The complete source materials for this paper, including all LaTeX source files, derivation documents, and revision history, are publicly available at: `https://github.com/denverdata/academic_InertialRelativity_STE_cases`.

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